



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Based Smart Microgrid Energy Management and Load Balancing Platform

Department of Computer Science and Engineering
CSA1017 — Software Engineering | CO2 Assessment Tool 1

NAME : MURALI KRISHNAN N

REG NO : 192524090

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for the AI-Based Smart Microgrid Energy Management and Load Balancing Platform. It is prepared to guide the design, development, and validation of a system that monitors, forecasts, and optimizes energy generation, storage, and consumption within a microgrid environment. The document serves as a reference for the development team, stakeholders, and evaluators to establish a shared and unambiguous understanding of system behavior before design and implementation begin.

1.2 Document Scope

This document covers the requirement analysis for the microgrid management platform, including the identification of actors, use cases, functional and non-functional requirements, data requirements, external interfaces, system features, assumptions, constraints, and a validation of the completeness and consistency of the captured requirements. It does not cover detailed system design, algorithm-level specifications, or implementation code.

1.3 Intended Audience

- Software architects and developers responsible for system design and implementation
- Energy and facility managers who will operate and rely on the platform
- Quality assurance engineers responsible for validation and testing
- Academic evaluators assessing the requirement analysis for CO2 Assessment Tool 1

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Microgrid	A localized group of electricity sources and loads that can operate connected to or independent of the main utility grid
DER	Distributed Energy Resource — small-scale generation units such as solar panels or wind turbines
BMS	Battery Management System — monitors and controls battery charge/discharge behavior
SCADA	Supervisory Control and Data Acquisition — industrial control system used for monitoring infrastructure

Term	Definition
AI	Artificial Intelligence — used here for demand forecasting and optimization
IoT	Internet of Things — network of smart meters and sensors feeding real-time data
SRS	Software Requirement Specification

2. Problem Description

2.1 Business Domain

The system operates in the energy and smart-infrastructure domain, specifically distributed energy management for microgrids deployed across campuses, industrial parks, and residential communities. These microgrids integrate renewable energy sources such as solar and wind alongside conventional supply and battery storage, and must be coordinated intelligently to remain reliable and cost-effective.

2.2 Stakeholders

Stakeholder	Interest / Role
Energy / Facility Manager	Oversees daily microgrid operations, reviews dashboards, and makes load-balancing decisions
System Administrator	Manages platform configuration, user access, and system health
Maintenance Technician	Responds to equipment faults and performs preventive maintenance on hardware assets
Utility Grid Operator	Coordinates power import/export transactions with the microgrid
Renewable Energy Providers	Supply intermittent solar/wind generation data to the platform
End Consumers (Campus/Residential Users)	Consume electricity and are indirectly affected by reliability and cost outcomes
Regulatory Bodies	Set compliance standards for grid interconnection, safety, and data handling

2.3 Existing Challenges

- Conventional energy management systems lack predictive capabilities and react only after imbalances occur
- Intermittent renewable generation (solar/wind) is difficult to forecast and integrate reliably
- Inefficient battery utilization increases operational costs and shortens battery lifespan
- Absence of real-time, unified dashboards makes it difficult to detect anomalies quickly
- Manual load balancing is slow, error-prone, and does not scale to larger microgrid deployments

2.4 System Objectives

- Provide real-time monitoring of energy generation, storage, and consumption
- Use AI-based forecasting to predict energy demand and generation trends
- Optimize battery charge/discharge cycles to minimize cost and extend battery life
- Balance electrical loads dynamically across available sources
- Improve overall energy reliability while reducing operational expenses
- Support sustainable, scalable power management across multiple microgrid sites

3. Scope of the System

The AI-Based Smart Microgrid Energy Management and Load Balancing Platform will provide an integrated software solution for monitoring and optimizing distributed energy resources. The system will interface with IoT smart meters, battery management systems, renewable energy sources, and the utility grid to collect real-time data, generate demand forecasts using AI models, and recommend or automatically execute load-balancing and battery-optimization actions.

In scope: real-time data acquisition, AI-based demand forecasting, battery optimization logic, load balancing engine, web/mobile dashboards, alerting and reporting, and role-based user management.

Out of scope: physical installation of IoT hardware, low-level firmware for smart meters, and direct control of third-party utility-grid infrastructure beyond the defined data-exchange interface.

4. Functional Requirements

The following table lists the functional requirements grouped by feature area, along with their priority levels.

ID	Category	Requirement Description	Priority
FR-01	Real-Time Monitoring	The system shall continuously collect and display real-time energy generation data from renewable sources (solar, wind).	High
FR-02	Real-Time Monitoring	The system shall continuously collect and display real-time energy consumption data across connected loads.	High
FR-03	Forecasting	The system shall use AI models to forecast short-term (hourly) and medium-term (daily) energy demand.	High
FR-04	Forecasting	The system shall forecast expected renewable generation based on historical and weather data.	High
FR-05	Battery Optimization	The system shall recommend or automatically execute optimal battery charge/discharge schedules.	High
FR-06	Battery Optimization	The system shall monitor battery state of charge (SoC) and state of health (SoH) continuously.	Medium
FR-07	Load Balancing	The system shall dynamically balance electrical loads across available generation, storage, and grid supply.	High
FR-08	Load Balancing	The system shall support automated load shedding for non-critical loads during supply shortages.	Medium
FR-09	Dashboard	The system shall provide a real-time operational dashboard showing generation, consumption, storage, and cost metrics.	High
FR-10	Alerts	The system shall generate configurable alerts/notifications for anomalies, faults, and threshold breaches.	High
FR-11	Reporting	The system shall generate periodic (daily/weekly/monthly) energy performance and cost-saving reports.	Medium

ID	Category	Requirement Description	Priority
FR-12	User Management	The system shall support role-based access control for administrators, managers, and technicians.	High
FR-13	Grid Interaction	The system shall record and manage electricity import/export transactions with the utility grid.	Medium
FR-14	Diagnostics	The system shall run periodic diagnostics on connected devices and flag faulty sensors or equipment.	Medium
FR-15	Historical Data	The system shall store historical generation, consumption, and battery data for trend analysis and AI model training.	High

5. Non-Functional Requirements

ID	Category	Requirement Description
NFR-01	Performance	The system shall process and display real-time sensor data with a maximum latency of 5 seconds.
NFR-02	Scalability	The system shall support scaling from a single microgrid site to multiple distributed sites without redesign.
NFR-03	Reliability	The system shall maintain 99.5% uptime for core monitoring and alerting functions.
NFR-04	Security	The system shall encrypt data in transit and at rest, and enforce role-based authentication for all users.
NFR-05	Usability	The dashboard shall be usable by non-technical facility staff with minimal training.
NFR-06	Maintainability	The system architecture shall support modular updates to AI models without downtime to core monitoring services.
NFR-07	Availability	Critical monitoring and alerting services shall remain available even if the forecasting module is temporarily offline.
NFR-08	Interoperability	The system shall support standard IoT communication protocols (e.g., MQTT, REST/HTTPS) for integration with third-party hardware.

6. Actors and Use Cases

6.1 Primary Actors

Actor	Description
Energy / Facility Manager	Monitors dashboards, reviews reports, configures alert thresholds, and makes operational decisions.
System Administrator	Manages user accounts, system configuration, and overall platform health.
Maintenance Technician	Responds to diagnostics and fault alerts; performs equipment maintenance.

6.2 Secondary Actors

Actor	Description
IoT Sensors / Smart Meters	Supply real-time generation and consumption data to the platform.
Battery Management System (BMS)	Provides battery status data and executes charge/discharge commands.
Weather Data Service	Supplies weather and irradiance data used for renewable-generation forecasting.
Utility Grid System	Exchanges electricity import/export data and pricing signals with the platform.

6.3 Use Case Summary

Actor	Use Case	Description
Energy/Facility Manager	View Operational Dashboard	View real-time generation, consumption, storage, and cost data.
Energy/Facility Manager	Configure Alerts & Thresholds	Define threshold values that trigger system alerts.
Energy/Facility Manager	Generate Energy Reports	Produce periodic performance and cost-saving reports.
Energy/Facility Manager	Monitor Real-Time Energy Generation	Track live output from renewable and conventional sources.
System Administrator	Manage User Accounts	Create, modify, and deactivate user accounts and roles.

Actor	Use Case	Description
System Administrator	Perform System Diagnostics	Run health checks on connected devices and services.
Maintenance Technician	Perform System Diagnostics	Investigate flagged equipment faults.
Maintenance Technician	Manage Renewable Source Integration	Onboard/configure new renewable generation units.
IoT Sensors/Smart Meters	Monitor Real-Time Energy Generation & Consumption	Feed live telemetry data into the platform.
Weather Data Service	Receive Weather & Irradiance Data	Supply forecast inputs for the AI forecasting engine.
Weather Data Service	Forecast Energy Demand (AI)	Support demand and generation forecasting models.
Utility Grid System	Exchange Grid Import/Export Data	Coordinate electricity transactions with the microgrid.
Battery Management System	Optimize Battery Storage Utilization	Execute recommended charge/discharge schedules.

6.4 Use Case Diagram

The diagram below illustrates the primary and secondary actors, the use cases within the system boundary, and their associations.

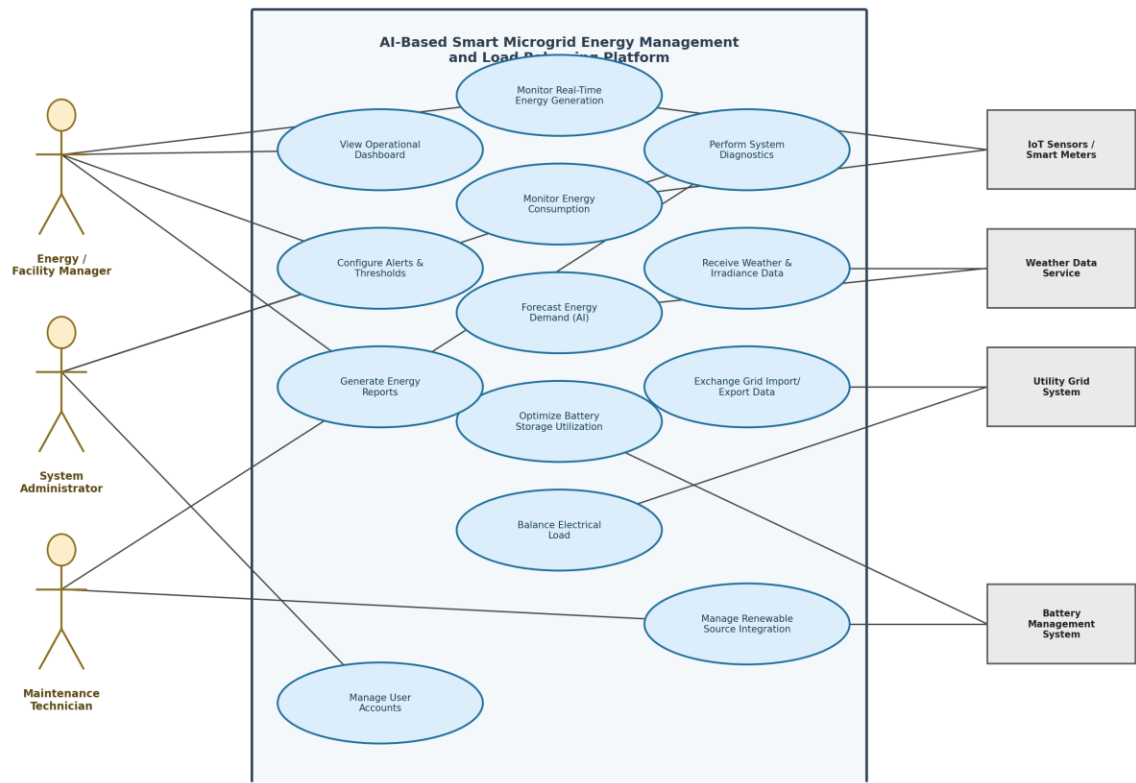


Figure 1: Use Case Diagram — AI-Based Smart Microgrid Energy Management and Load Balancing Platform

6.5 Detailed Use Case Descriptions

UC-03 — Forecast Energy Demand (AI)

Use Case ID	UC-03
Use Case Name	Forecast Energy Demand (AI)
Primary Actor(s)	Energy/Facility Manager (primary), Weather Data Service (secondary)
Preconditions	Historical consumption data and current weather data are available to the system.
Main Flow	1. System retrieves historical generation and consumption data. 2. System requests current weather/irradiance data from the Weather Data Service. 3. AI forecasting engine generates short-term and medium-term demand and generation forecasts. 4. Forecast results are displayed on the operational dashboard. 5. System flags forecasts with low confidence for manager review.

Alternate / Exception Flow	• If weather data is unavailable, the system uses the last known forecast and displays a data-staleness warning.
Postconditions	Updated demand and generation forecasts are available to the load balancing and battery optimization modules.

UC-05 — Optimize Battery Storage Utilization

Use Case ID	UC-05
Use Case Name	Optimize Battery Storage Utilization
Primary Actor(s)	Battery Management System (secondary), Energy/Facility Manager (primary)
Preconditions	Battery SoC/SoH data is available; demand forecast has been generated.
Main Flow	1. System retrieves the latest demand and generation forecast. 2. System evaluates current battery SoC and SoH. 3. Optimization logic calculates the ideal charge/discharge schedule. 4. System sends the schedule to the Battery Management System for execution. 5. System logs the executed action and resulting battery status.
Alternate / Exception Flow	• If battery SoH falls below a safe threshold, the system restricts discharge and raises a maintenance alert.
Postconditions	Battery charge/discharge schedule is updated and logged for reporting.

UC-07 — Balance Electrical Load

Use Case ID	UC-07
Use Case Name	Balance Electrical Load
Primary Actor(s)	Energy/Facility Manager (primary), Utility Grid System (secondary)
Preconditions	Real-time generation, consumption, and battery status data are available.
Main Flow	1. System continuously compares real-time generation with consumption.

	2. System determines whether a supply deficit or surplus exists. 3. If a deficit exists, system draws from battery storage or the utility grid based on cost and priority rules. 4. If a surplus exists, system either charges the battery or exports power to the grid. 5. System updates the dashboard with the current load-balancing status.
Alternate / Exception Flow	• If both battery and grid supply are insufficient, the system triggers automated load shedding for non-critical loads (FR-08).
Postconditions	Electrical load remains balanced within defined operational limits.

7. Data Requirements

Data Entity	Description	Nature
Energy Generation Data	Real-time and historical output from solar, wind, and conventional sources	Time-series, per source (kWh)
Energy Consumption Data	Real-time and historical load data from connected consumption points	Time-series, per load (kWh)
Battery Status Data	State of charge, state of health, charge/discharge rate	Time-series (% cycles)
Weather Data	Solar irradiance, wind speed, temperature	Time-series, external feed
Grid Transaction Data	Import/export volumes and pricing with the utility grid	Time-series, transactional
User & Role Data	Account credentials, roles, and permission levels	Relational, access-controlled
Alerts & Event Logs	Threshold breaches, faults, and system diagnostic events	Event log, timestamped

8. External Interface Requirements

8.1 User Interfaces

- Web-based dashboard for facility managers and administrators, accessible via standard browsers.
- Mobile-responsive view for on-the-go monitoring and alert notifications.

8.2 Hardware Interfaces

- IoT smart meters and sensors connected via standard field protocols (e.g., Modbus, Zigbee gateways).
- Battery Management System (BMS) interface for reading status and issuing charge/discharge commands.

8.3 Software Interfaces

- REST/HTTPS APIs for integration with cloud services and the AI forecasting engine.
- Third-party weather data API for irradiance and meteorological forecasts.
- Utility grid data-exchange interface for import/export and pricing signals.

8.4 Communication Interfaces

- MQTT protocol for lightweight, real-time telemetry from IoT devices.
- Secure HTTPS/TLS channels for all external API communication.

9. System Features

Feature	Description	Priority
Real-Time Monitoring Dashboard	Unified view of generation, consumption, battery status, and cost metrics.	High
AI-Based Demand Forecasting	Short- and medium-term forecasts for demand and renewable generation.	High
Battery Optimization Engine	Recommends/executes optimal charge-discharge schedules.	High
Load Balancing Engine	Dynamically allocates load across generation, storage, and grid supply.	High
Alert & Notification System	Configurable, real-time alerts for anomalies and threshold breaches.	High
Reporting & Analytics	Periodic performance and cost-saving reports with historical trends.	Medium

Feature	Description	Priority
Role-Based Access Control	Differentiated access levels for managers, administrators, and technicians.	Medium

10. Assumptions and Constraints

10.1 Assumptions

- Reliable internet/network connectivity is available at all microgrid sites.
- IoT smart meters and sensors are properly installed, calibrated, and periodically maintained.
- Sufficient historical energy data exists or will be collected to train and validate AI forecasting models.
- End users (managers, administrators) receive basic training on operating the dashboard.
- Connected hardware supports standard communication protocols (MQTT, Modbus, REST).

10.2 Constraints

Type	Constraint
Technical	The system must integrate with existing legacy SCADA and metering hardware at some sites, limiting protocol choices.
Operational	Core monitoring and alerting services must operate 24/7 with minimal planned downtime.
Economic	IoT sensor deployment and cloud infrastructure costs must remain within the project's allocated budget.
Legal	The system must comply with regional data-privacy regulations and grid-interconnection standards.
Time	Requirement analysis and initial platform design must be completed within the academic assessment timeline.

11. Requirement Validation

11.1 Completeness Check

All stakeholder concerns identified in Section 2.2 are addressed: monitoring and reporting needs (Facility Manager), access/system management (Administrator), fault handling (Technician), and data exchange needs (Grid Operator, Weather Service, Battery Management System). Each system objective in Section 2.4 maps to at least one functional requirement in Section 4.

11.2 Consistency Check

Functional requirements were cross-checked against the use cases in Section 6 to confirm that every use case is supported by at least one functional requirement, and that no requirement contradicts another (e.g., automated load shedding under FR-08 is consistent with the load balancing behavior described in UC-07).

11.3 Ambiguity and Redundancy Check

- Quantitative thresholds (e.g., latency, uptime) were added to non-functional requirements to reduce ambiguity — see NFR-01 and NFR-03.
- Overlapping monitoring requirements (FR-01/FR-02) were kept separate since generation and consumption are measured independently, avoiding redundant merged requirements.
- No conflicting priority assignments were found across the functional requirement set.

11.4 Suggested Improvements

- Define explicit data-retention periods for historical data (Section 7) to support both compliance and AI model retraining needs.
- Add a formal requirement for AI forecast accuracy thresholds to make forecasting quality measurable and testable.
- Introduce a disaster-recovery/backup requirement to strengthen system reliability beyond NFR-03.
- Consider a phased rollout plan (single-site pilot before multi-site scaling) to validate scalability assumptions (NFR-02) incrementally.